DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY

Neural Network Lab - Assessment Report

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Student Information

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Overall Result

26/28 (92.9%) Grade: A

Score by Section

Section	Score	Pct
Activation Functions	5/5	100.0%
Weight Initialization	3/3	100.0%
Loss Functions	3/3	100.0%
Optimizers	4/4	100.0%
Backpropagation	3/3	100.0%
Regularization	2/4	50.0%
Batch Normalization	2/2	100.0%
Architecture Design	4/4	100.0%

Questions to Review

Q21. [Regularization]

L2 regularization adds lambda multiplied by the sum of squared weights to the loss. What effect does this have?

*L2 adds lambda*W to each gradient update, decaying weights toward 0 but never exactly 0. Also called weight decay.*

Q22. [Regularization]

Which regularization setup produces the SMALLEST train/val loss gap?

L2 + Dropout combines weight penalty and random deactivation, providing stronger regularization than either alone.